Multiplication in $\mathbb{N}$

1. 6×7
2. 8×24
3. 35×16
4. 142×28
5. 325×412
6. $4 \times 15 \times 3$
7. $25 \times 16 \times 4$
8. $8 \times 15 \times 25 \times 4$
9. An auditorium has 38 rows of seats, with each row containing 24 seats. What is the total seating capacity of the auditorium?
10. A warehouse receives a shipment of 125 pallets. Each pallet contains 48 boxes, and each box holds 16 individual items. What is the total number of items in the shipment?

Solutions

1. **Answer:** 42
   Explanation: Basic single-digit multiplication fact: $6 \times 7 = 42$.

2. **Answer:** 192

$$\begin{array}{r} {\scriptstyle 3} \\ 24 \\ \times\ \ 8 \\ \hline 192 \end{array}$$

3. **Answer:** 560

$$\begin{array}{r} {\scriptstyle 3} \\ 35 \\ \times 16 \\ \hline 210 \\ 350 \\ \hline 560 \end{array}$$

4. **Answer:** $3,976$

$$\begin{array}{r} {\scriptstyle 3\,1} \\ 1428 \\ \times\ \ 28 \\ \hline 1136 \\ 2840 \\ \hline 3976 \end{array}$$

5. **Answer:** $133,900$

$$\begin{array}{r} {\scriptstyle 1\,1} \\ 325 \\ \times\ 412 \\ \hline 650 \\ 3250 \\ 130000 \\ \hline {\scriptstyle 1} \\ 133900 \end{array}$$

6. **Answer:** 180
   Explanation: First compute $4 \times 15 = 60$, then multiply 60×3:

$$\begin{array}{r} {\scriptstyle 2} \\ 15 \\ \times\ \ 4 \\ \hline 60 \end{array} \qquad \begin{array}{r} 60 \\ \times\ \ 3 \\ \hline 180 \end{array}$$

7. **Answer:** $1,600$
   Explanation: Group factors to simplify ($25 \times 4 = 100$), then calculate 100×16:

$$\begin{array}{r} {\scriptstyle 2} \\ 25 \\ \times\ \ 4 \\ \hline 100 \end{array} \qquad \begin{array}{r} 100 \\ \times\ \ 16 \\ \hline 600 \\ 1000 \\ \hline 1600 \end{array}$$

8. **Answer:** $12,000$
   Explanation: Pair factors strategically ($25 \times 4 = 100$ and $8 \times 15 = 120$), then calculate 120×100:

$$\begin{array}{r} {\scriptstyle 4} \\ 15 \\ \times\ \ 8 \\ \hline 120 \end{array} \qquad \begin{array}{r} 120 \\ \times\ \ 100 \\ \hline 0 \\ 00 \\ 12000 \\ \hline 12000 \end{array}$$

9. **Answer:** 912 **seats**

$$\begin{array}{r} {\scriptstyle 3} \\ 38 \\ \times 24 \\ \hline 152 \\ 760 \\ \hline {\scriptstyle 1} \\ 912 \end{array}$$

10. **Answer:** $96,000$ **items**
    Explanation: Multiply $125 \times 48 \times 16$. Evaluating $125 \times 16 = 2,000$ first, then $2,000 \times 48$:

$$\begin{array}{r} {\scriptstyle 1\,3} \\ 125 \\ \times\ \ 16 \\ \hline 750 \\ 1250 \\ \hline 2000 \end{array} \qquad \begin{array}{r} 2000 \\ \times\ \ 48 \\ \hline 160000 \\ 800000 \\ \hline {\scriptstyle 1} \\ 960000 \end{array}$$